

PHD PROGRAM

Applied Data Science & Artificial Intelligence



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Explainable Artificial Intelligence



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PDPs and ICE



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Summary of the lecture

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In the second part we will briefly discuss the metrics used in the literature to evaluate XAI approaches.

Main idea

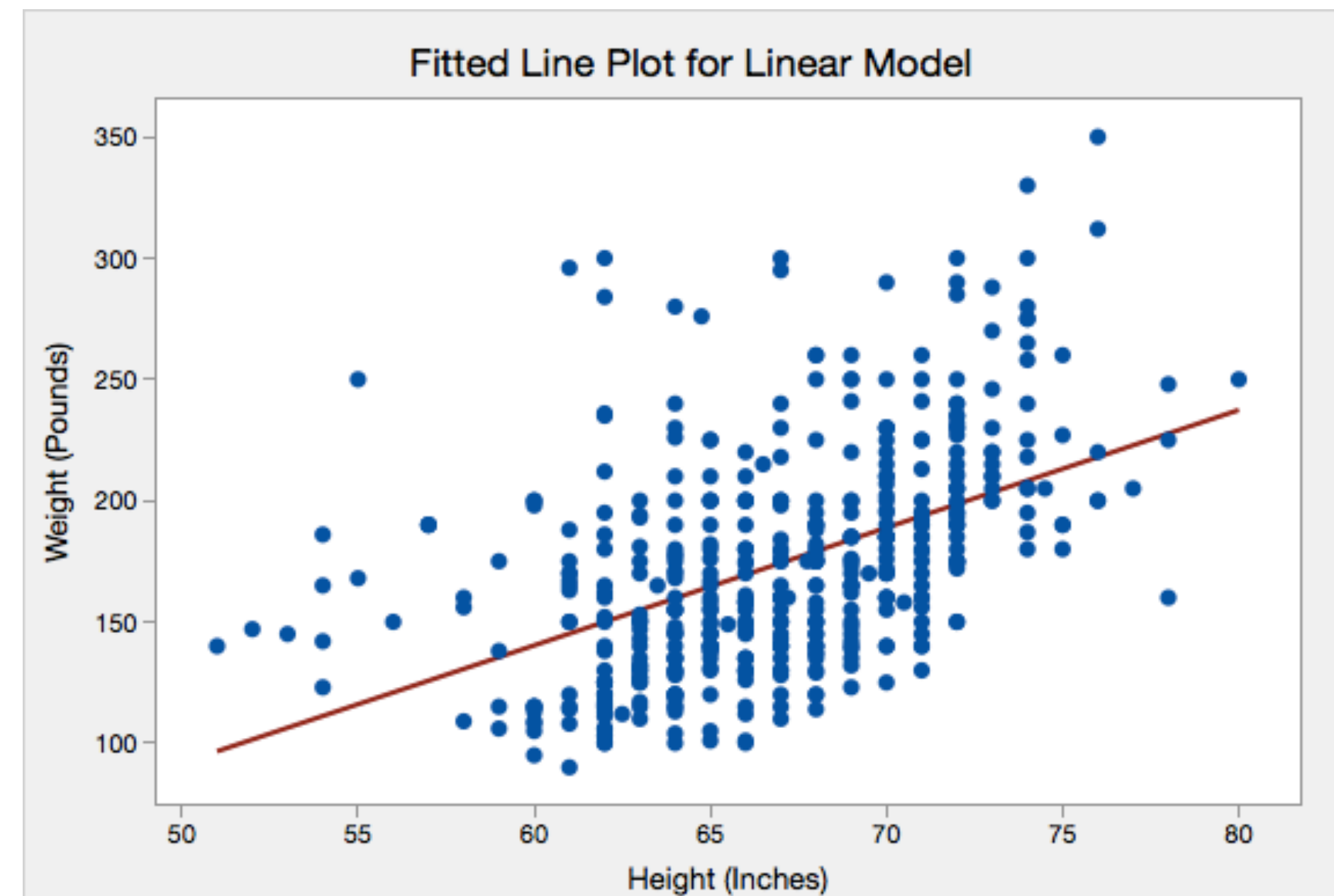
XAI aims at explaining model predictions.

What is an explanation?

Given a model, an *intuitive* idea of an explanation could be quantifying the effect of each input variable on the output of the model.

In the following we will see approaches based on this idea. Since these methods require only input data X , a generic model $F(X)$ and its input y without any specific knowledge of $F(\cdot)$ they are called ***model-agnostic methods***.

Partial Dependency Plots - Intro

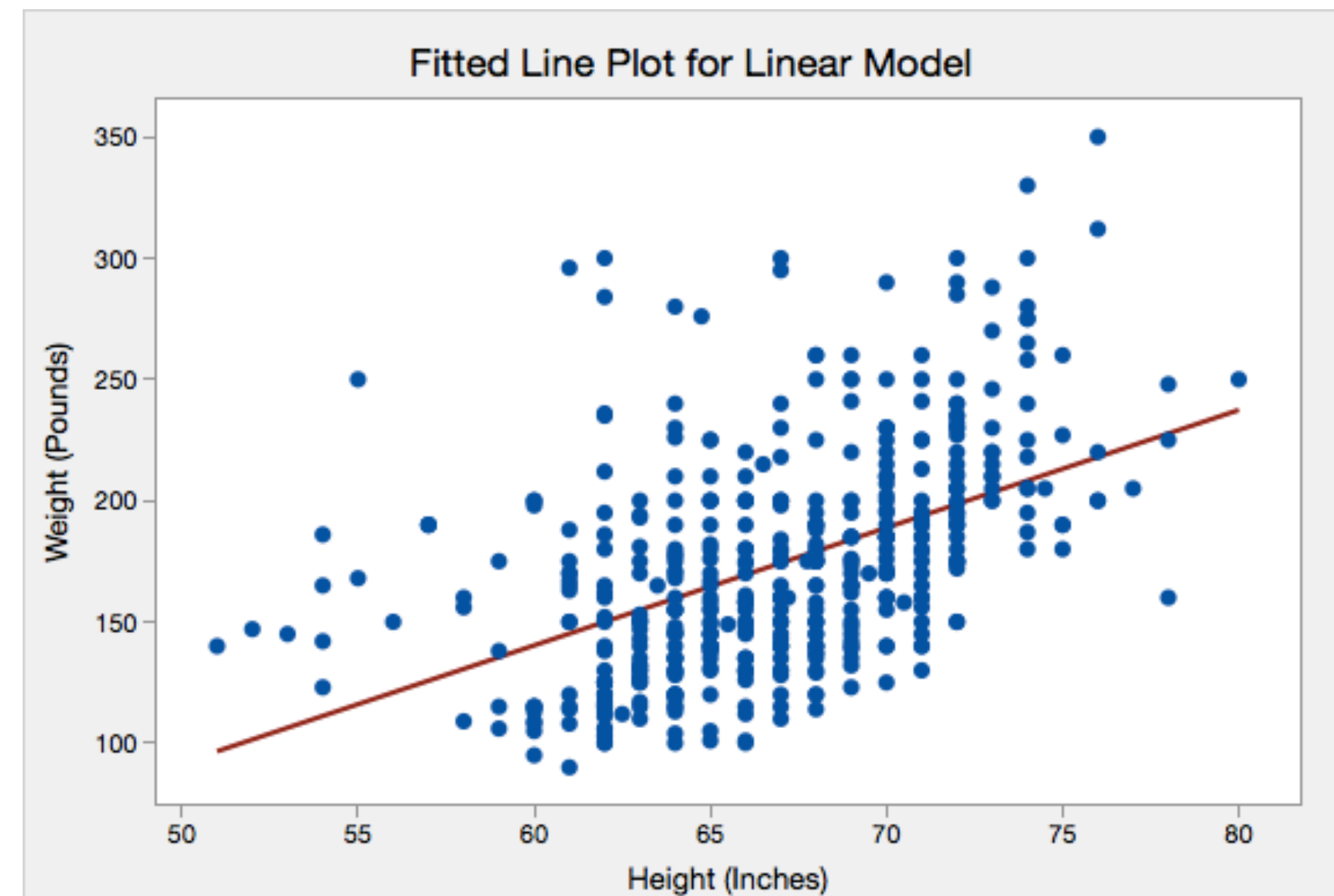


<https://online.stat.psu.edu/stat200/sites/stat200/files/inline-images/HeightWeight.png>

Partial Dependency Plots - Intro

Visualization is a powerful tool for interpretability.

Consider a univariate regression model $y = f(x)$, then we can have an intuitive understanding of the effect of x on y by plotting one against the other.



<https://online.stat.psu.edu/stat200/sites/stat200/files/inline-images/HeightWeight.png>

Partial Dependency Plots – ctd.

[1] J. H. Friedman. “Greedy function approximation: A gradient boosting machine.” 2001

Partial Dependency Plots – ctd.

Problem: how can we visualize the effect of variables in a high dimensional setting?

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Partial Dependency Plots – ctd.

Problem: how can we visualize the effect of variables in a high dimensional setting?

Let $X = (x_1, x_2, \dots, x_n)$ be a n -dimensional dataset and let $F(X)$ be a model with output y . Consider now an l -dimensional subset of the data $X_l = (x_1, \dots, x_l) \subset X$. We are interested in visualizing the effect of X_l on the output y .

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Friedman *et al.* [1] propose to do so by plotting the **partial dependence** of y from X_l .

[1] J. H. Friedman. "Greedy function approximation: A gradient boosting machine." 2001

Partial Dependency Plots - definition

Partial Dependency Plots - definition

Let X_{-l} be the complement of the subset of interest X_l , so that $X_l \cup X_{-l} = X$.
Then, the partial dependence of the model $F()$ with respect to X_l calculated in a point $x_l \in X_l$ is defined as follows

$$pd_{X_l}(x_l) = E_{X_{-l}}[F(x_l, X_{-l})] = \int F(x_l, x_{-l}) p(x_{-l}) dx_{-l}$$

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$$pd_{X_l}(x_l) = E_{X_{-l}}[F(x_l, X_{-l})] = \int F(x_l, x_{-l}) p(x_{-l}) dx_{-l}$$

Which in practice is approximated by

$$\overline{F_{X_l}(x_l)} = \frac{1}{n} \sum_{i=0}^n F(x_l, x_{i, -l})$$

Where $x_{i, -l}$ are the samples from X_{-l} available in the training set.

Partial Dependency Plots - explanation

Partial Dependency Plots - explanation

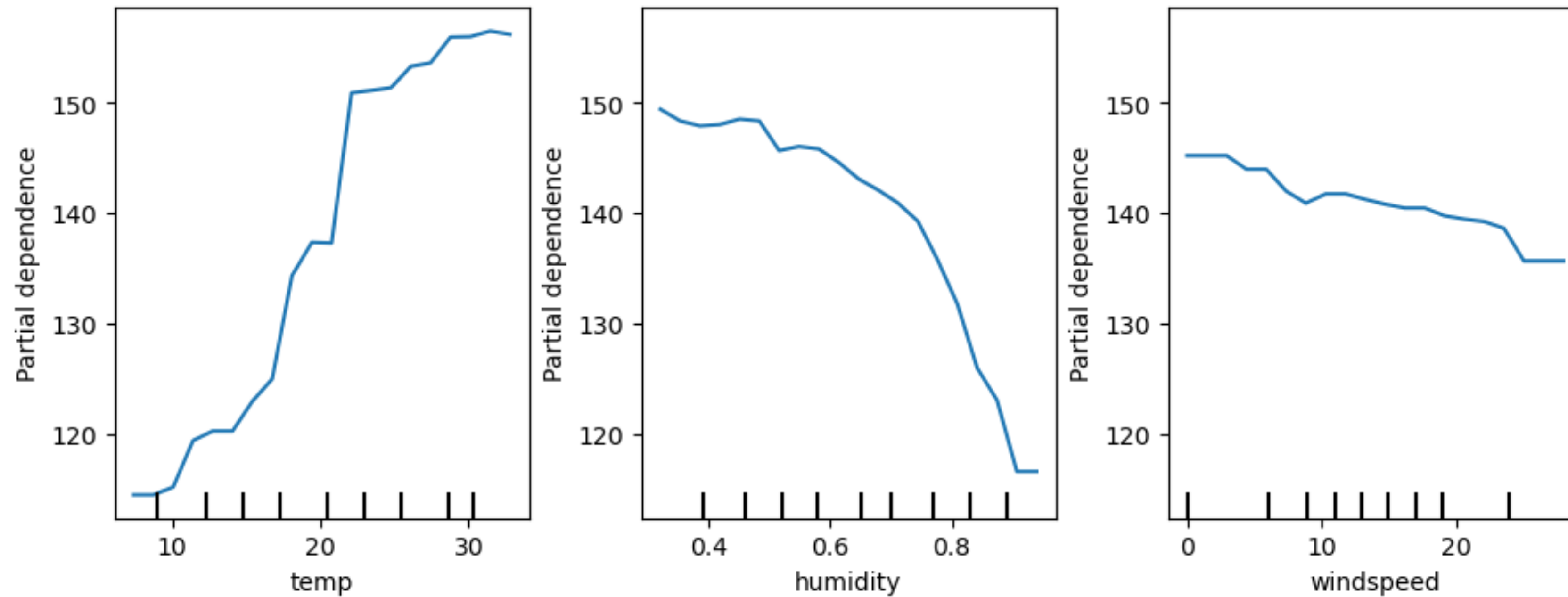
The approximated partial dependence is representative of the effect of the subset of interest on the average output taken with respect to the complementary set.

In PDPs this value is plotted to visualize the marginal effect of changes in subset of variables of interest on the average output value of the model.

In practice, often $l = 1$ is the preferred choice, so that the subset of interest consists of a single variable.

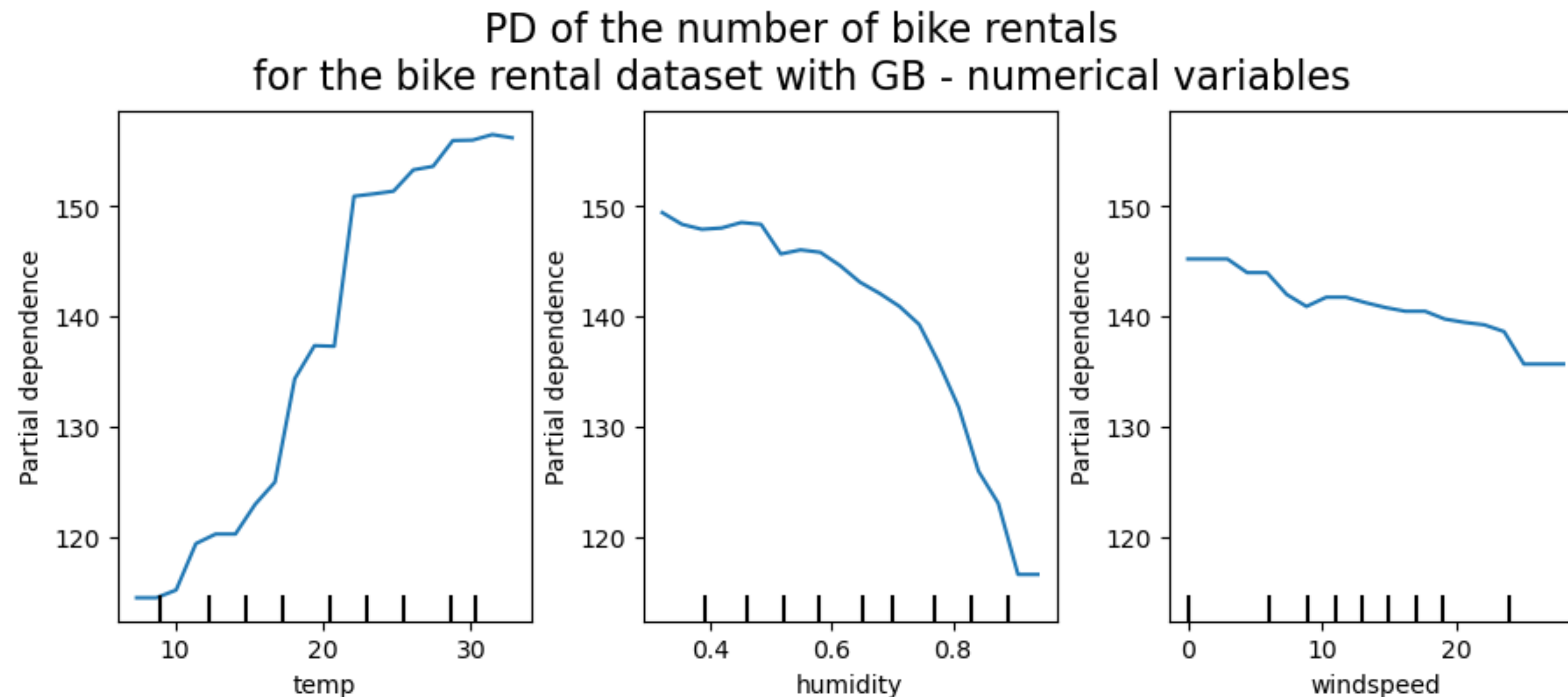
Partial Dependency Plots - Examples

PD of the number of bike rentals
for the bike rental dataset with GB - numerical variables



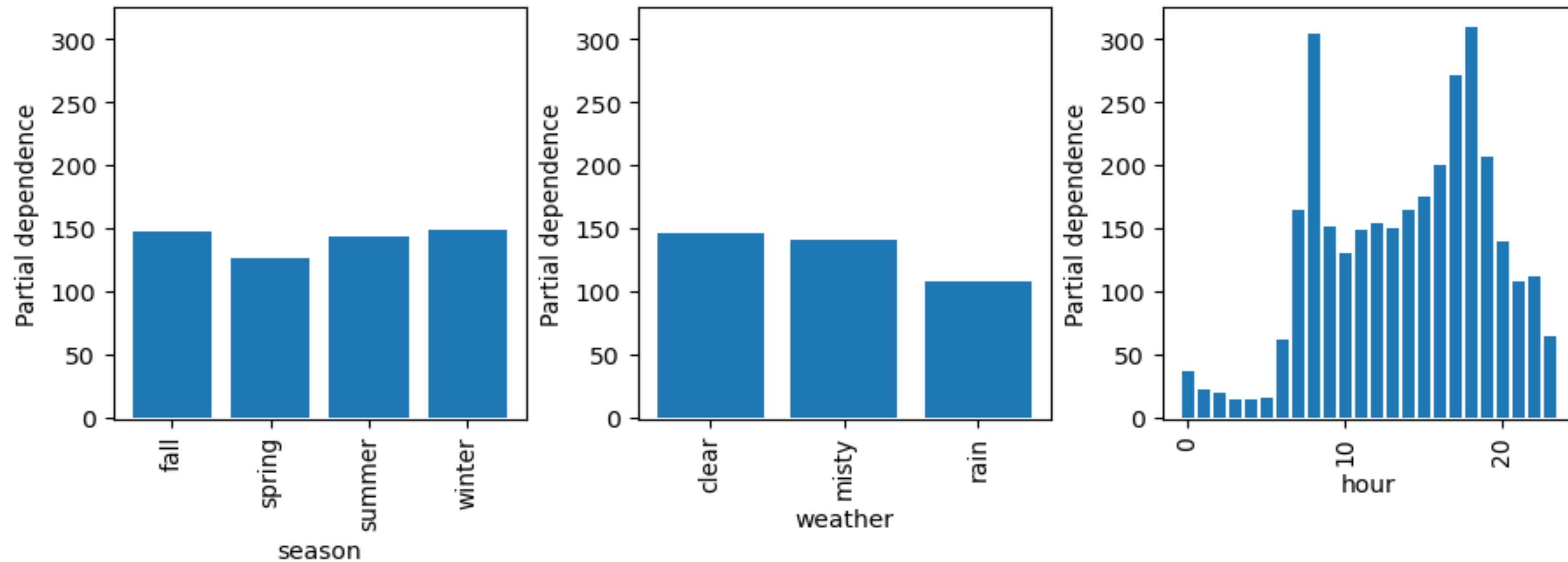
Partial Dependency Plots - Examples

We now have a first look at the PDPs for a Gradient Boosting model that we will take as an example during the laboratory. First, we look at the plot for three numerical variable with continuous support.



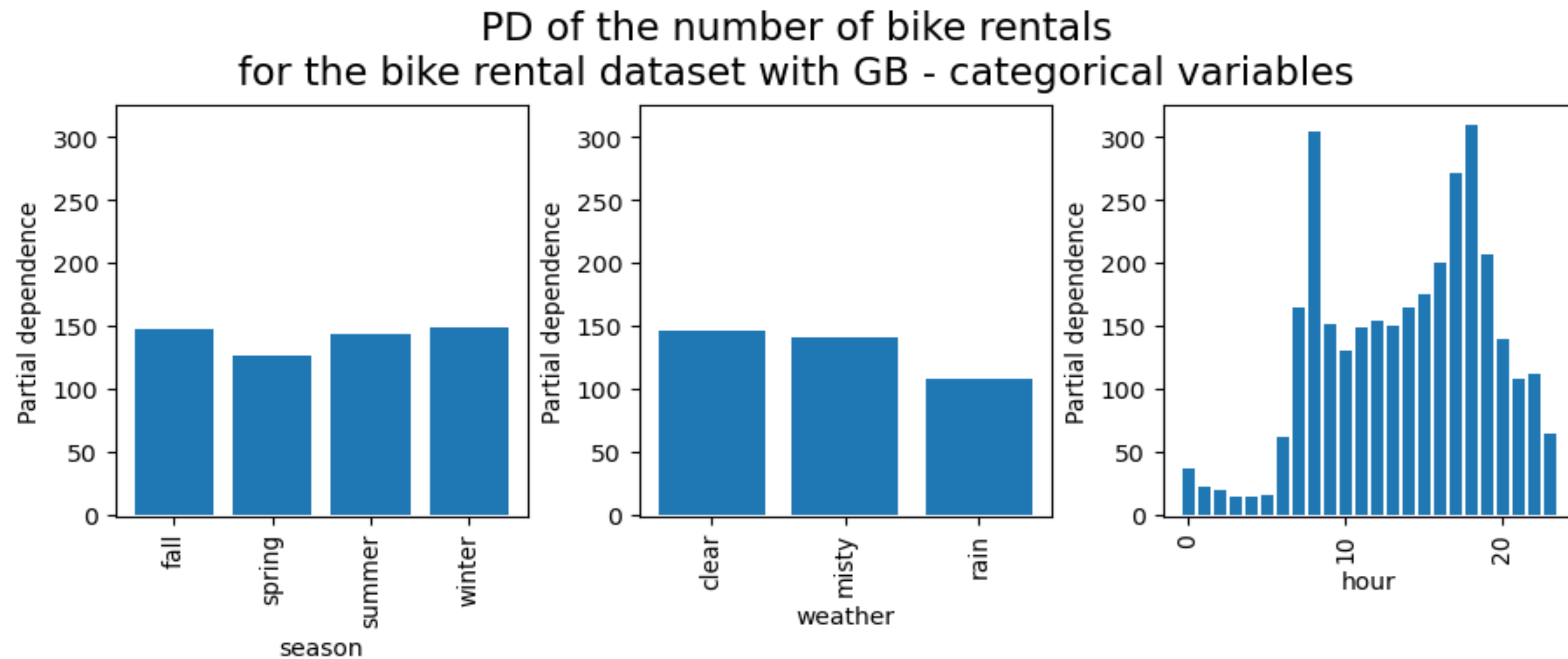
Partial Dependency Plots - Examples

PD of the number of bike rentals
for the bike rental dataset with GB - categorical variables



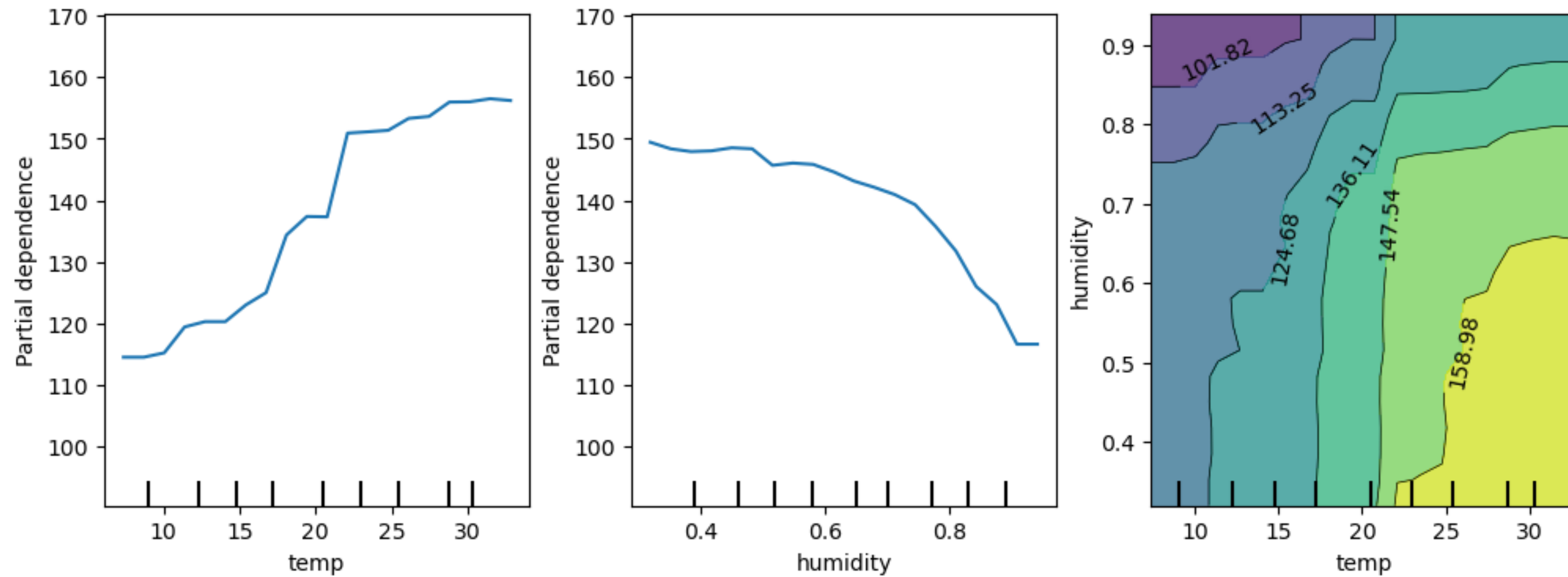
Partial Dependency Plots - Examples

Here we have PDPs calculated on variables with discrete support.



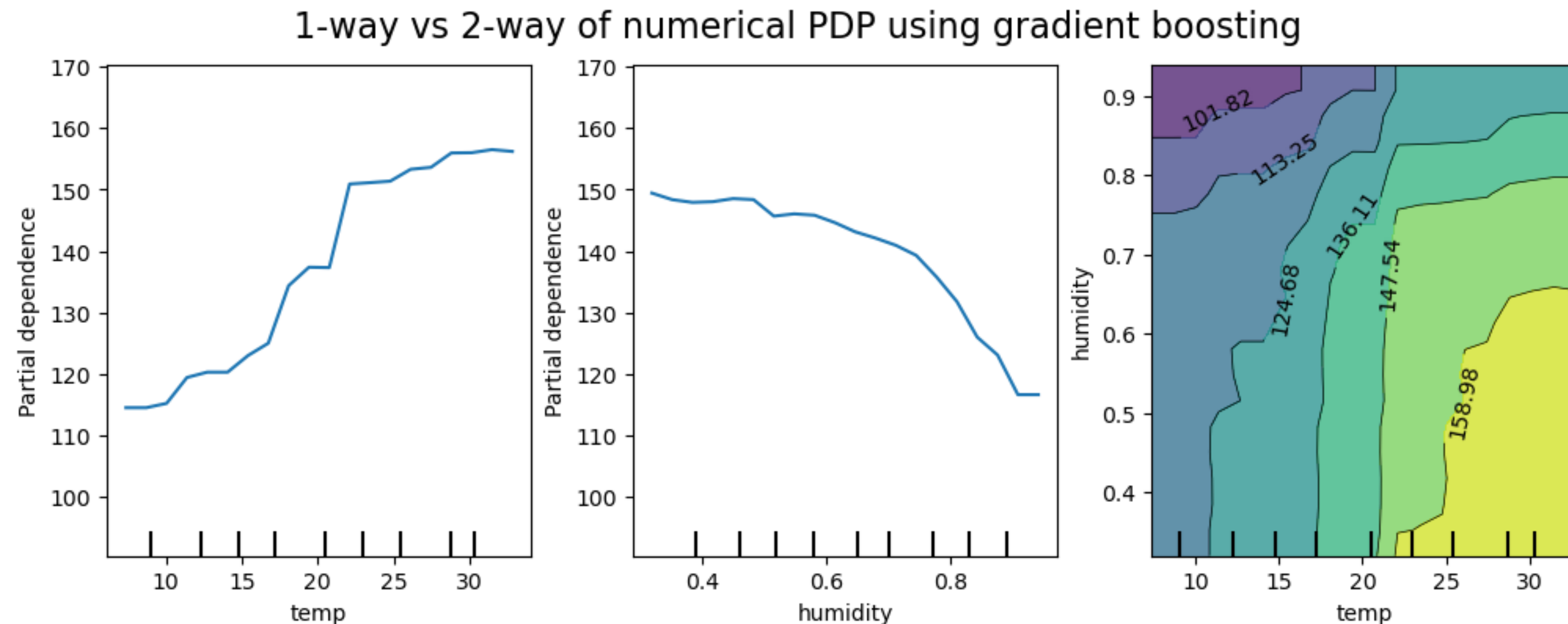
Partial Dependency Plots - Examples

1-way vs 2-way of numerical PDP using gradient boosting

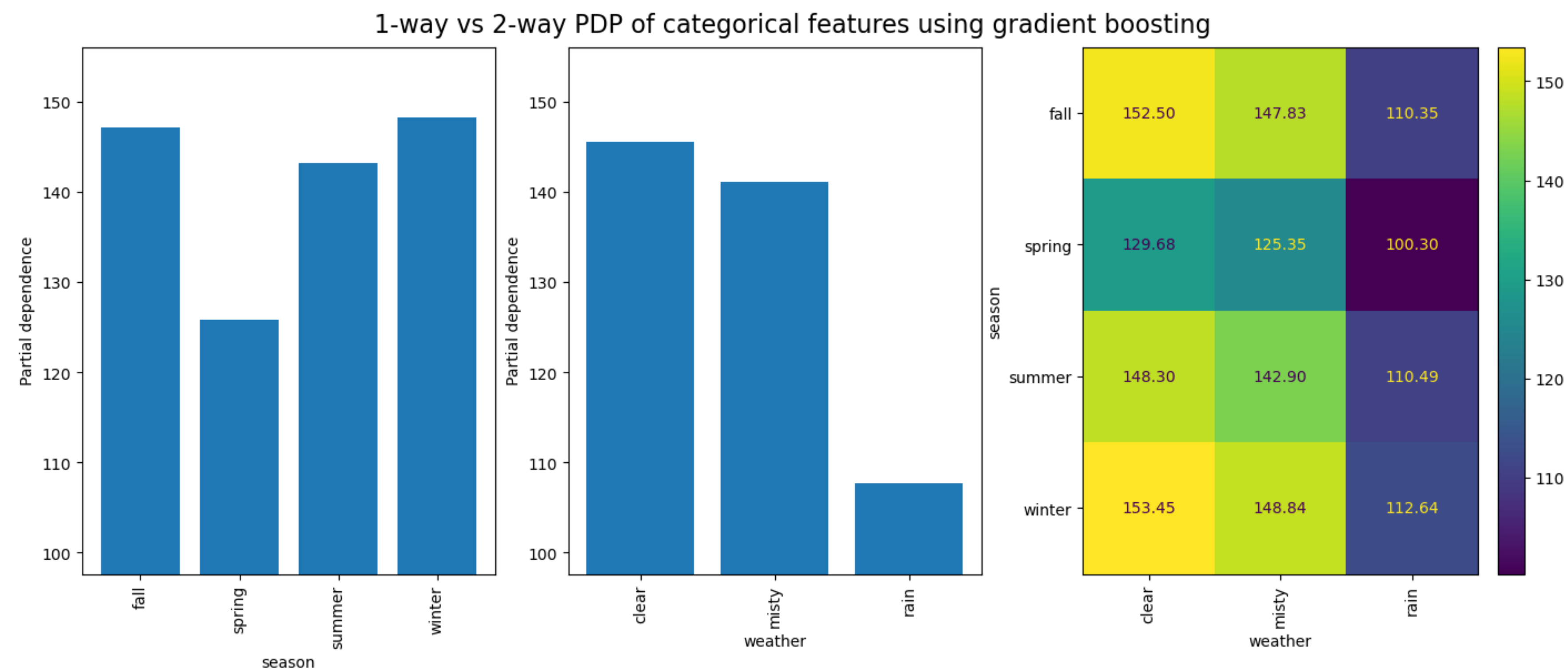


Partial Dependency Plots - Examples

For the same data and model, we consider a PDP with $l = 2$ for continuous variables. By doing so, we are able to visualize interactions between the variables in the subset.

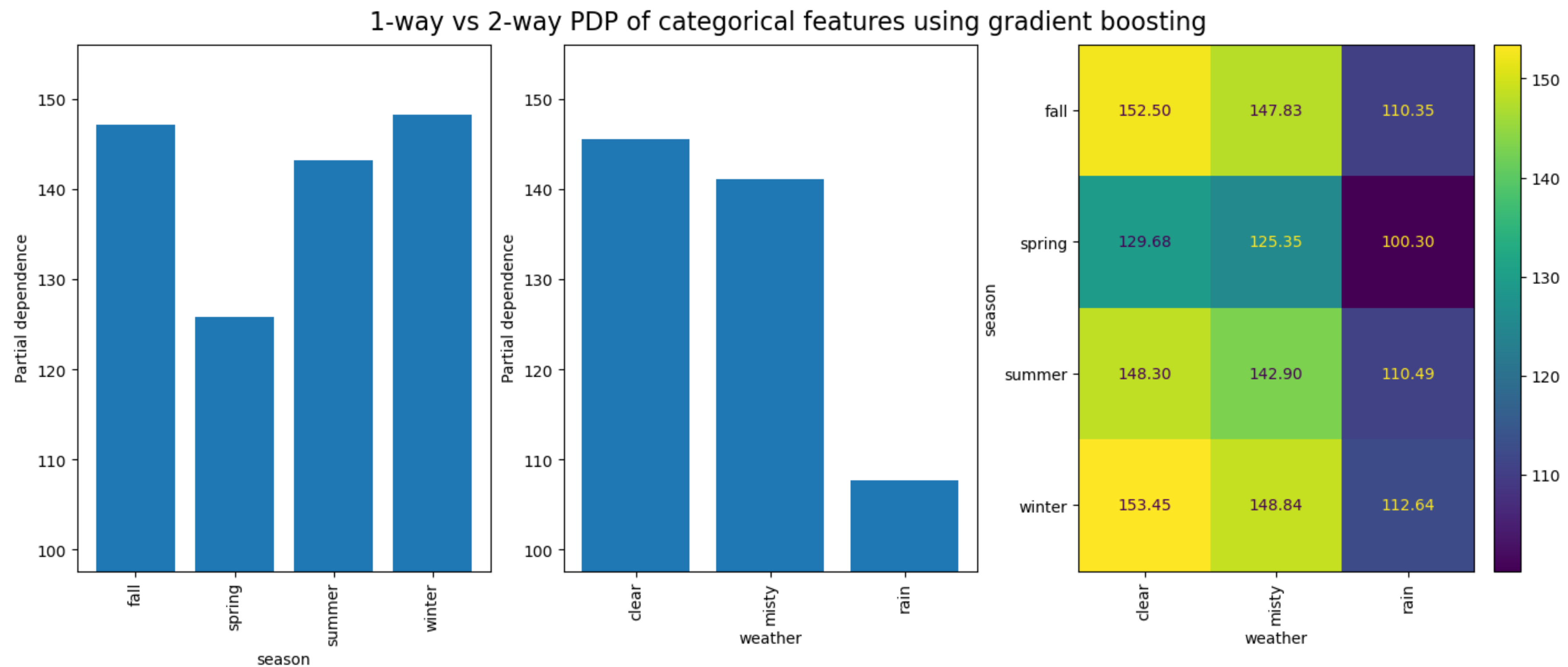


Partial Dependency Plots - Examples



Partial Dependency Plots - Examples

Similarly, we can visualize a PDP for a two-dimensional subset of categorical variables.



Partial Dependency Plots - comments

Partial Dependency Plots - comments

For both continuous and discrete variables, PDPs show how changes in the variable(s) of interest affects the average output value.

Moreover, when the subset of interest is not one dimensional, PDPs can be used to visualize interactions in the subset.

However, PDPs do not account for interactions between the subset and its complement.

Partial Dependency Plots - weaknesses

[2] A. Goldstein et al. “Peeking Inside the Black Box: Visualizing Statistical Learning with Plots of Individual Conditional Expectation.” 2014.

Partial Dependency Plots - weaknesses

Recall now how the partial dependence is computed:

$$\overline{F_{X_l}(x_l)} = \frac{1}{n} \sum_{i=0}^n F(x_l, x_{i, -l})$$

Since for each x_l the partial dependence it's computed averaging w.r.t. all values of $x_{i, -l}$ in the training set, PDPs do not account for heterogeneous relationships in the data.

To address this issue, in 2015 *Goldstein et al.* [2] proposed an extension of PDPs which they called **Individual Conditional Expectation** plots (ICEs for short).

[2] A. Goldstein et al. "Peeking Inside the Black Box: Visualizing Statistical Learning with Plots of Individual Conditional Expectation." 2014.

Individual Conditional Expectation plots - idea

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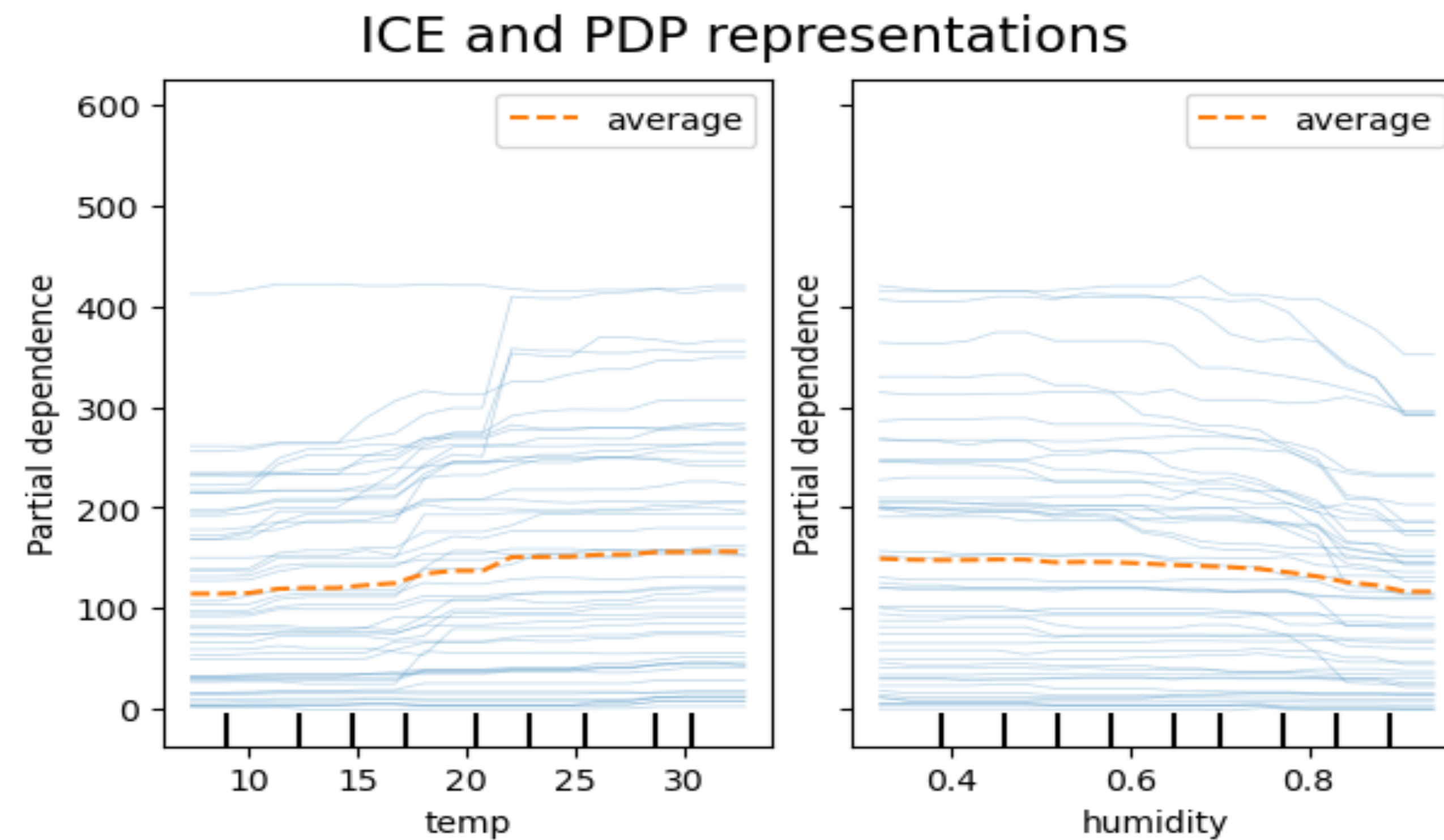
In their most simple formulation ICEs are a straightforward extension of the PDPs.

As mentioned before, the inability of PDPs to account for interactions in the variables it's due to the averaging used to compute the partial dependency.

In ICEs, the marginal effect of the subset of interest X_l is visualized for each single point $x_{i,-l}$ of the complement set. Thus, ICE curves are given by

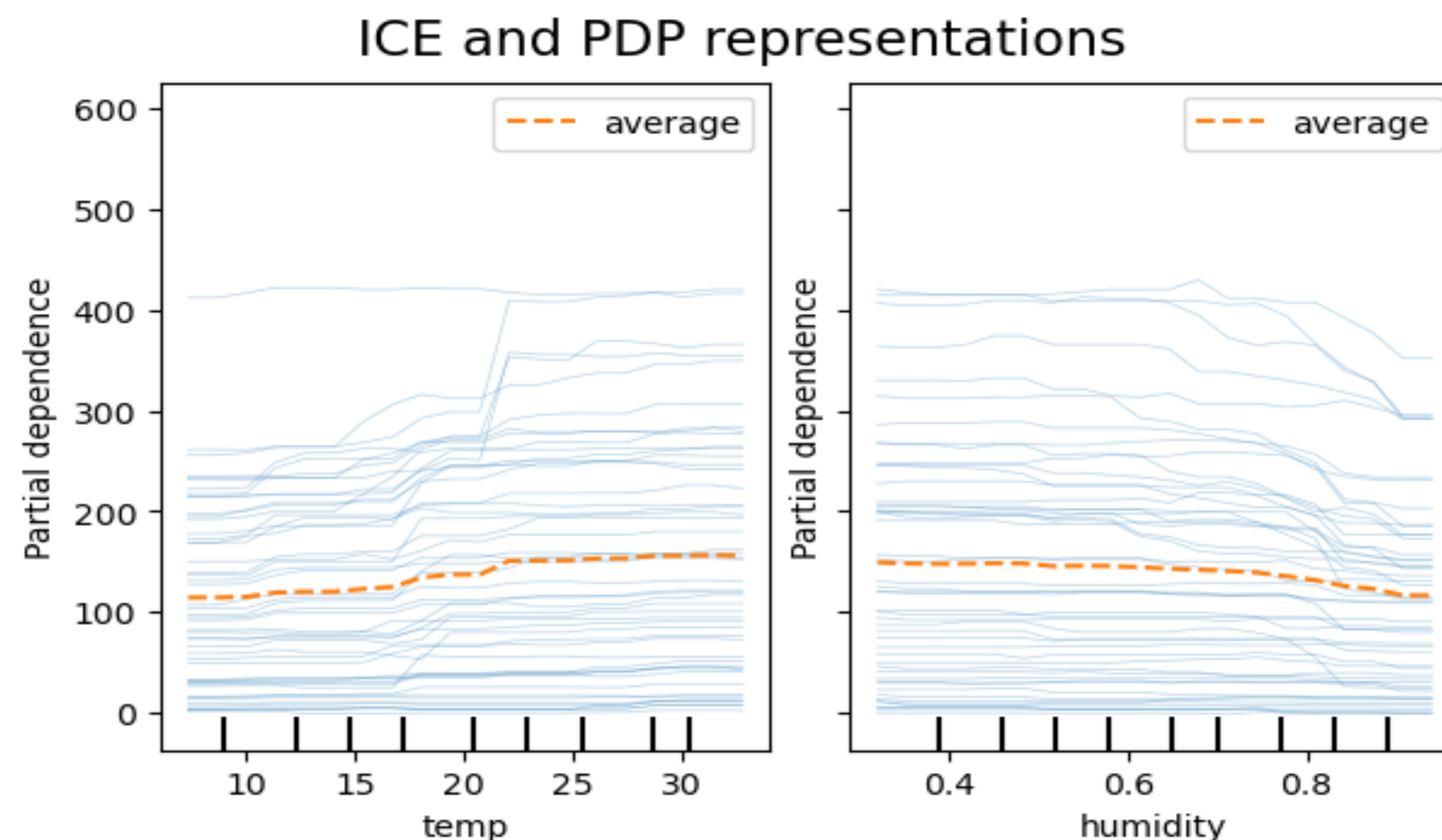
$$F_i(x_l) = F(x_l, x_{i,-l}) \text{ for all } x_{i,-l} \text{ in the training set}$$

Individual Conditional Expectation plots



Individual Conditional Expectation plots

Here we show the ICE computed for two different one-dimensional subsets. In orange it is highlighted the average value of the partial dependency (i.e. the correspondent PDP)



Individual Conditional Expectation plots – extensions 1

Individual Conditional Expectation plots – extensions 1

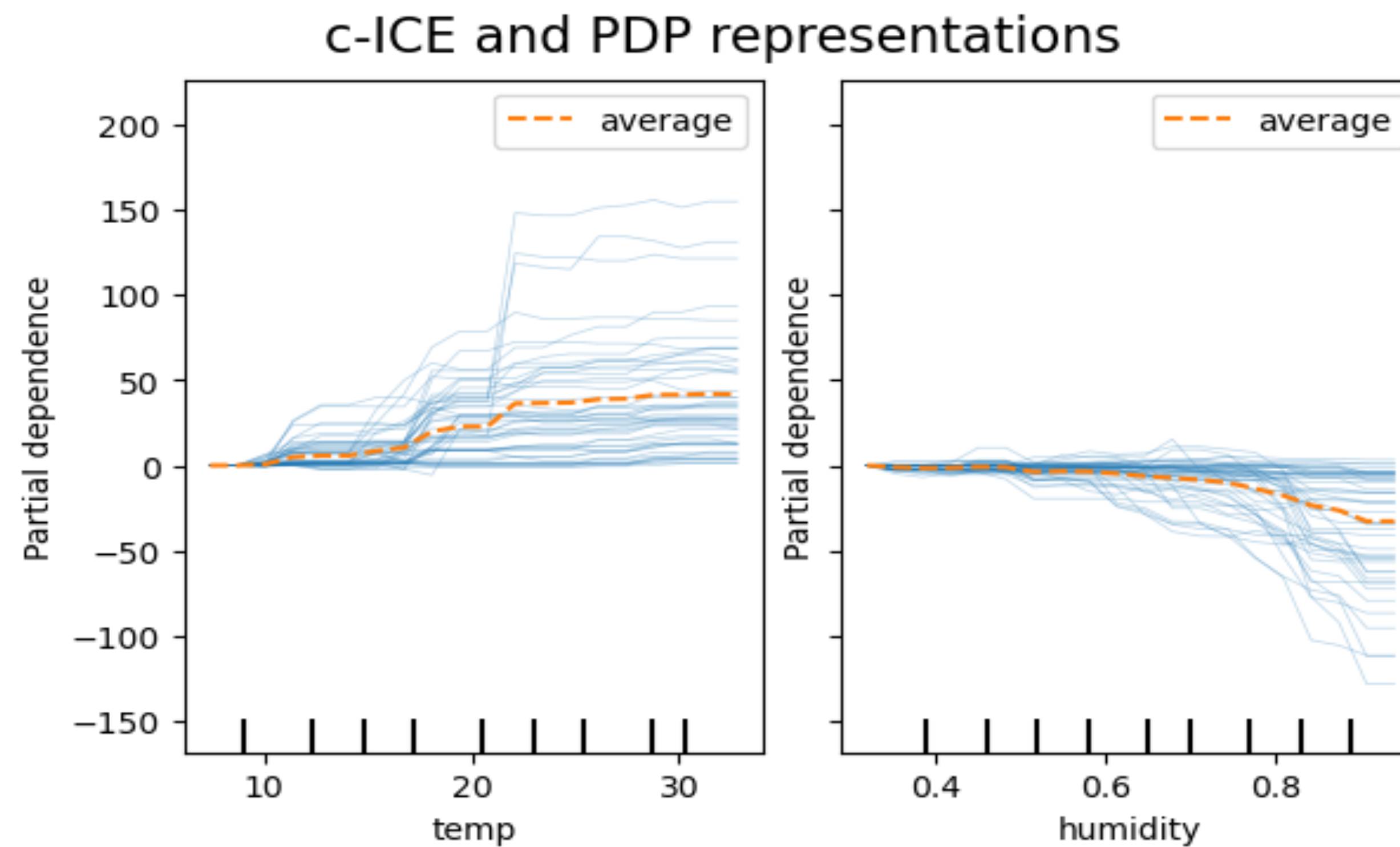
To improve visualization and human interpretability, the ICE plot can be modified into *the centered ICE* plot (or **c-ICE**).

First a baseline point $x^* \in X_l$ is chosen (commonly the minimum or the maximum in the subset of interest). Then, the c-ICE is computed by taking

$$F_i(x_l) = F(x_l, x_{i,-l}) - 1F(x^*, x_{i,-l}) \quad \text{for all } x_{i,-l} \text{ in the training set}$$

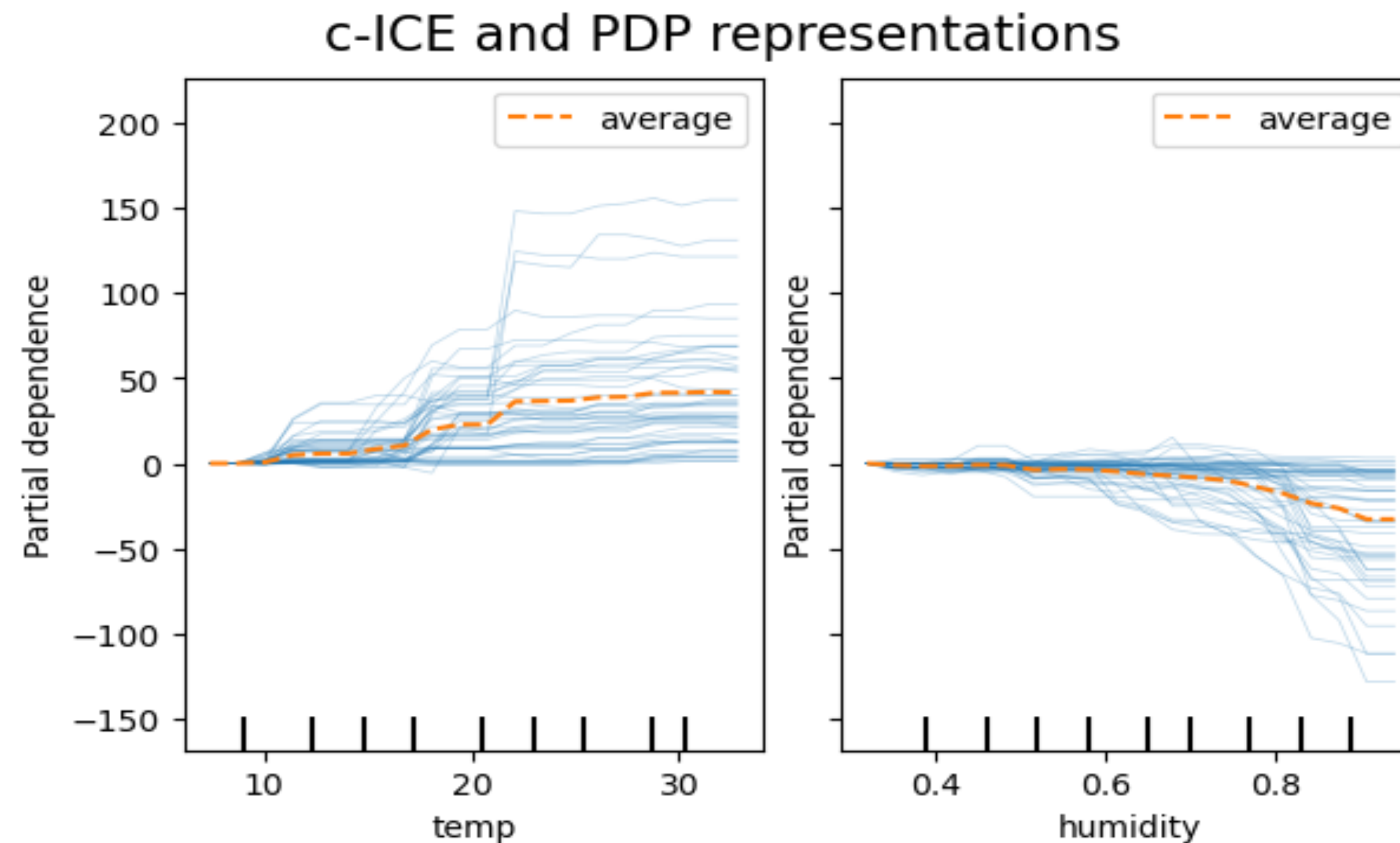
Often, interactions among the variables become more apparent when looking at the c-ICE.

Individual Conditional Expectation plots



Individual Conditional Expectation plots

As done for the simple ICE, we plot the c-ICE along with the centered average.



Individual Conditional Expectation plots – extensions 2

Individual Conditional Expectation plots – extensions 2

Another tool developed on the idea of ICE is the so called *derivative ICE* or simply **d-ICE**. These plots represent the derivative of the fitted model w.r.t. the subset of interest X_l .

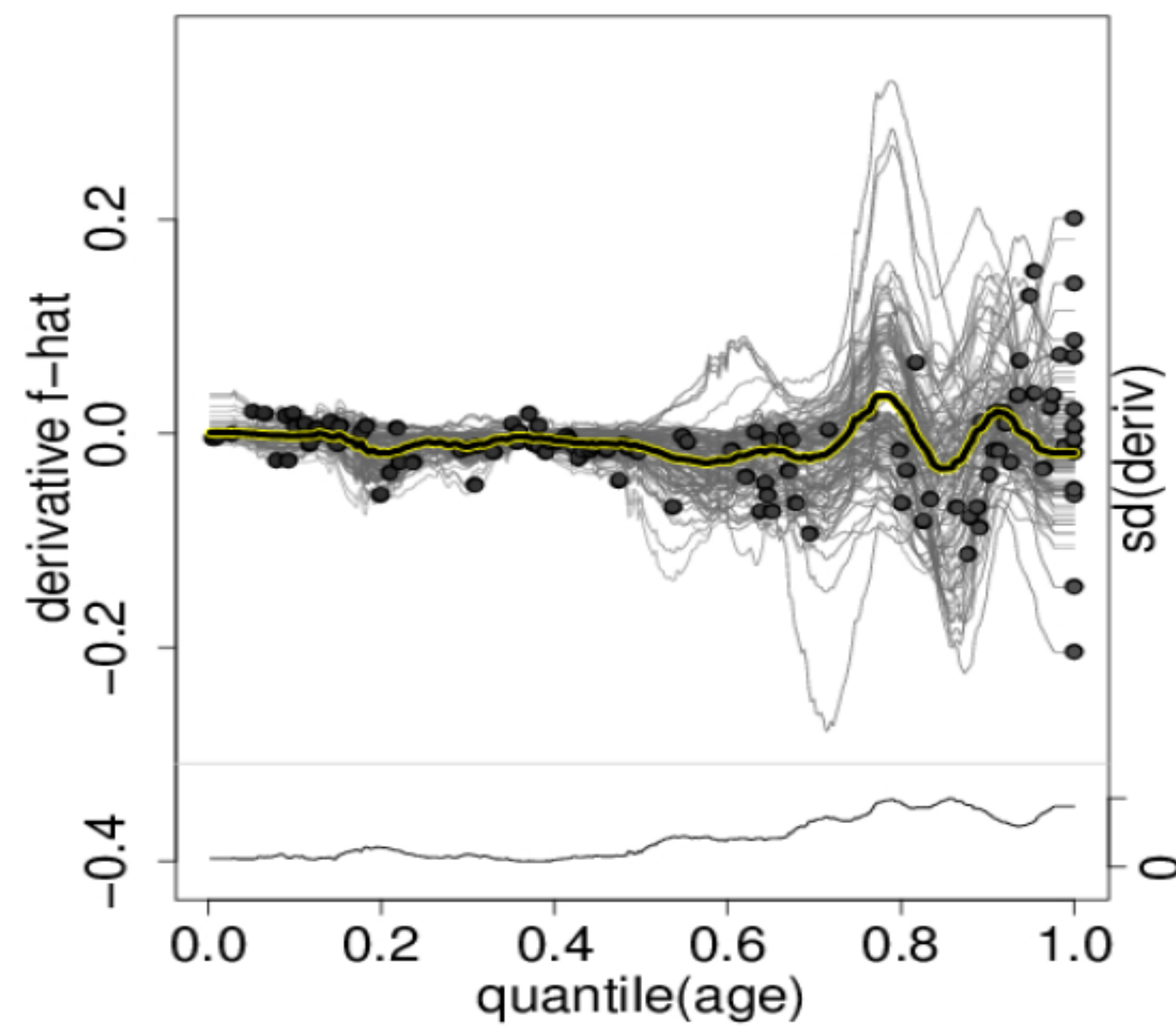
The rationale behind this version of the plot, is that when X_l does not interact with X_{-l} , the model can then be written as follows:

$$F(X) = F(X_l, X_{-l}) = g(X_l) + h(X_{-l})$$

And thus the derivative would be $\partial_{X_l} F(X) = g'(X_l)$. This means that, without interactions, the individual partial derivatives should have the same form for all instances. If they differ, it is due to interactions and it becomes visible in the d-ICE plot.

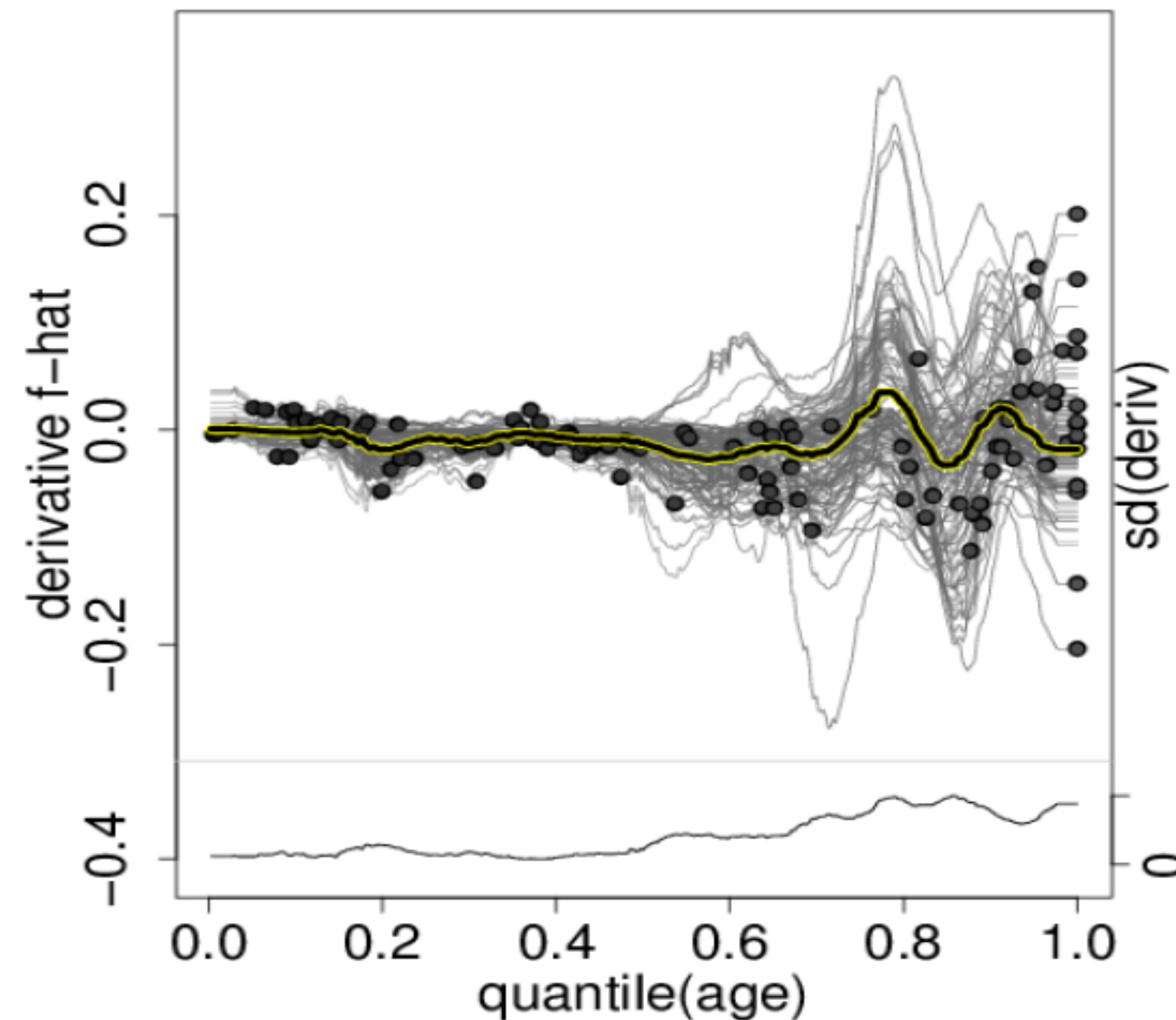
Usually, an estimate of the individual partial derivatives is plotted.

Derivative Individual Conditional Expectation plots



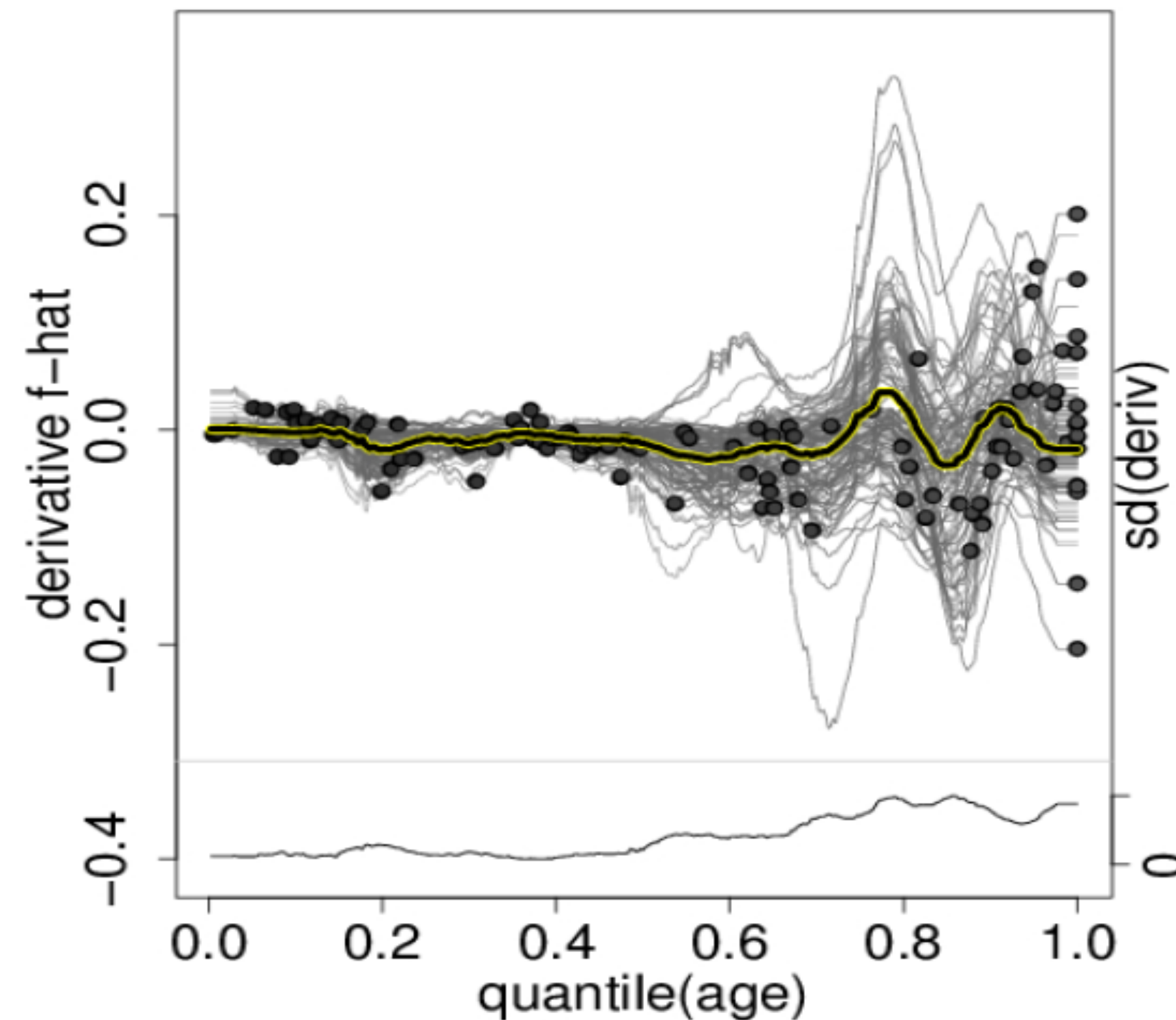
Derivative Individual Conditional Expectation plots

Example of the d-ICE plots for the *Boston Housing Data*. The model is a Random Forest used to predict a census tract's median home price using features of the census tract itself. The derivatives are taken w.r.t. the variable “*age quantile*”. Below, standard deviation of the derivative is also plotted.



Derivative Individual Conditional Expectation plots

Example of the d-ICE plots for the *Boston Housing Data*. The model is a Random Forest used to predict a census tract's median home price using features of the census tract itself. The derivatives are taken w.r.t. the variable “*age quantile*”. Below, standard deviation of the derivative is also plotted.



The plot suggests that after the 0.5 quantile of age there are interactions between age and the other variables, as the derivatives become less stable. This is also confirmed by the standard deviation plot.

Uses of ICEs : additivity assessment

Uses of ICEs : additivity assessment

ICEs can be used to evaluate to which extent the considered model fits the hypothesis of an additive model, *i.e.* that X_l does not interact with X_{-l} , so that the model can be written as:

$$F(X) = F(X_l, X_{-l}) = g(X_l) + h(X_{-l})$$

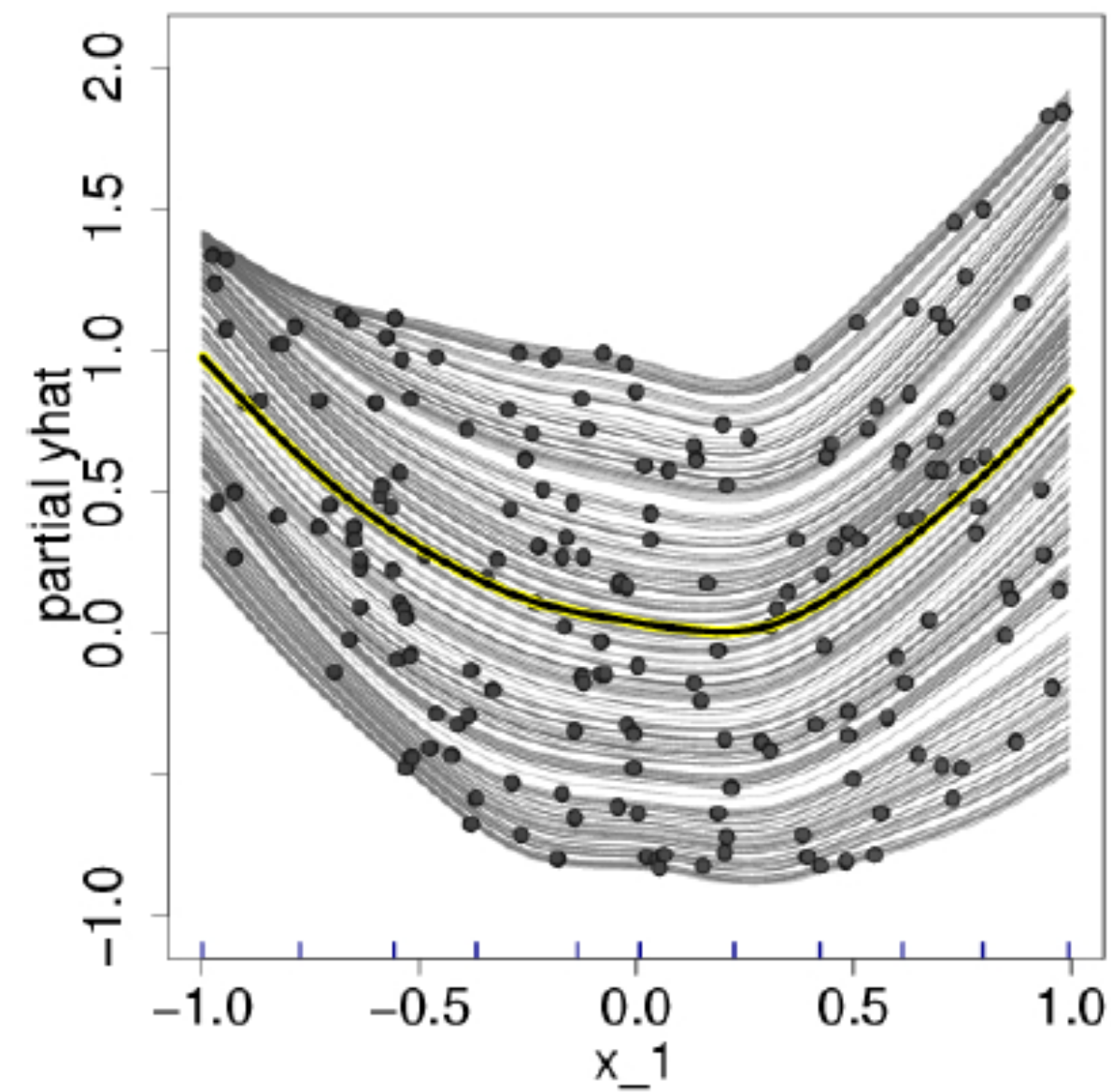
In a purely additive model, its ICE lines would display a set of curves equivalent in shape but differing in level, and similarly the d-ICE should show little to no variance in the derivatives.

In the following we will consider as an example a model fitted on data generated by the following additive structure

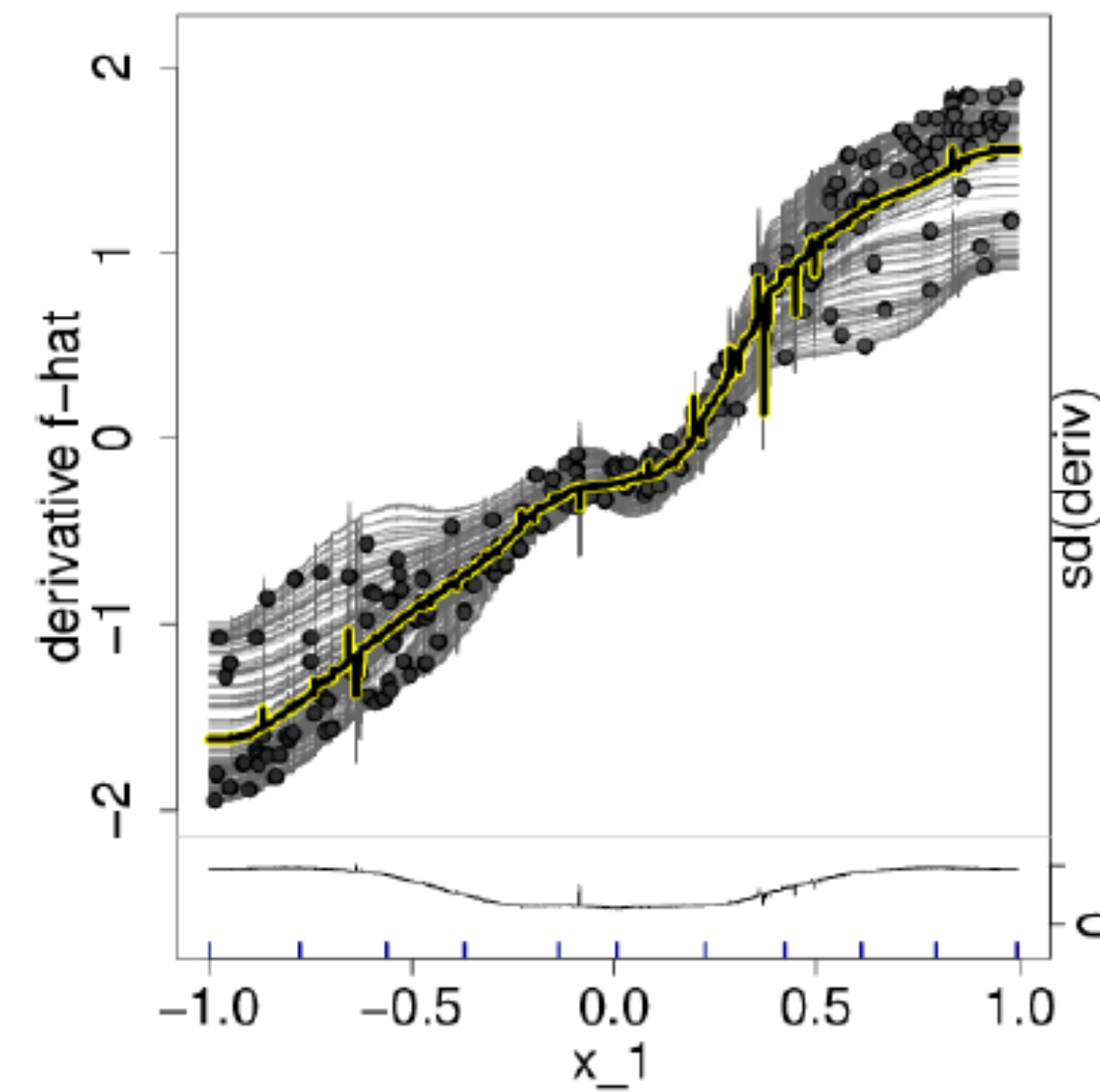
$$Y = X_1^2 + X_2 + \varepsilon$$

Where X_1, X_2 are i.i.d. variables and ε is a noise parameter.

Uses of ICEs : additivity assessment



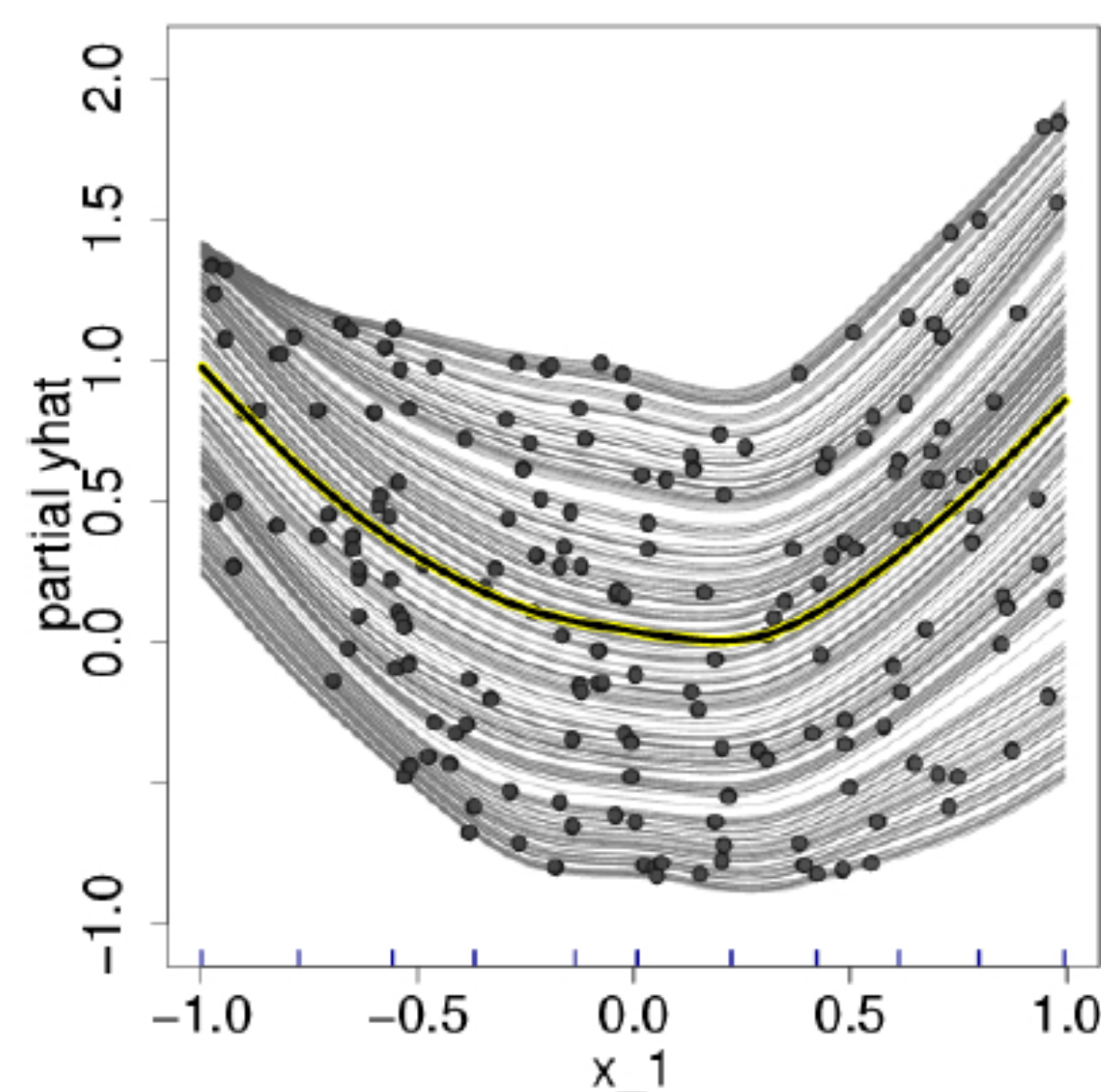
(a) ICE



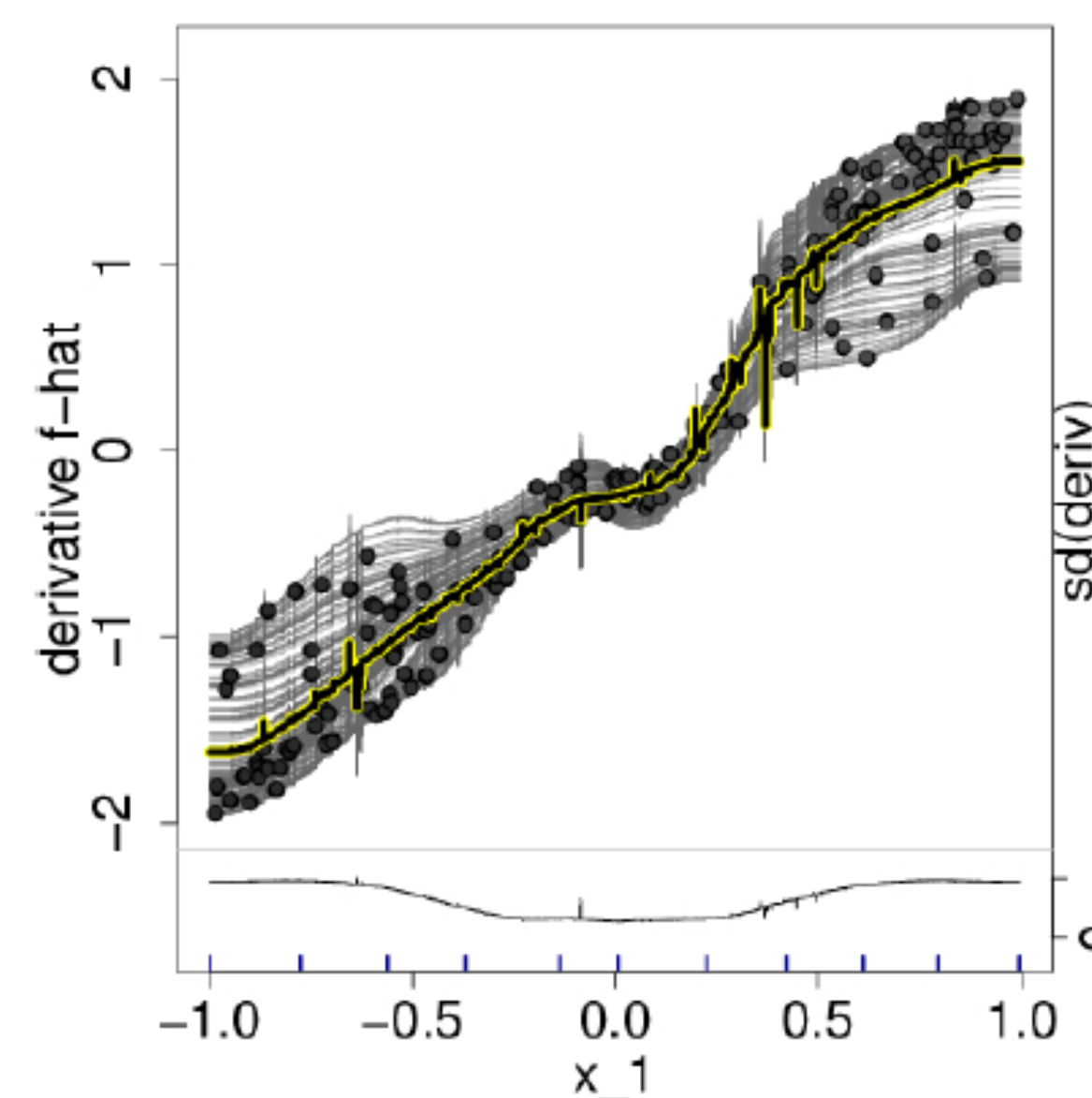
(b) d-ICE

Uses of ICEs : additivity assessment

As we have defined the generated data the model fitted should capture only a minimal interaction between X_1 and X_2 , due to the noise. To prove this, we fit a generalized additive model (GAM) on the data and show both the ICE and d-ICE plots. Both plots indeed confirm the additive hypothesis. This assessment would have been impossible with PDP alone.



(a) ICE



(b) d-ICE

Uses of ICEs : extrapolation detection

$$Y = 10X_1^2 + \mathbb{1}_{X_2 \geq 0} + \mathcal{E},$$

$$\mathcal{E} \stackrel{iid}{\sim} \mathcal{N}(0, .1^2), \quad \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} \sim \begin{cases} \text{U}(-1, 0), \text{U}(-1, 0) & \text{w.p. } \frac{1}{3} \\ \text{U}(0, 1), \text{U}(-1, 0) & \text{w.p. } \frac{1}{3} \\ \text{U}(-1, 0), \text{U}(0, 1) & \text{w.p. } \frac{1}{3} \\ \text{U}(0, 1), \text{U}(0, 1) & \text{w.p. } 0. \end{cases}$$

Uses of ICEs : extrapolation detection

As the number of variables in a model increases, the task of forecasting a value for a new point x_* using model F becomes an extrapolation task (*i.e.* the new point lies outside the convex hull of the points in the training set) rather than an interpolation one. This type of task has a great amount of uncertainty associated, and a black-box model precludes us from gaining any insight into what the extrapolation might look like.

The ICE plots can be used to detect how the model might extrapolate.

Consider data generated by the following model:

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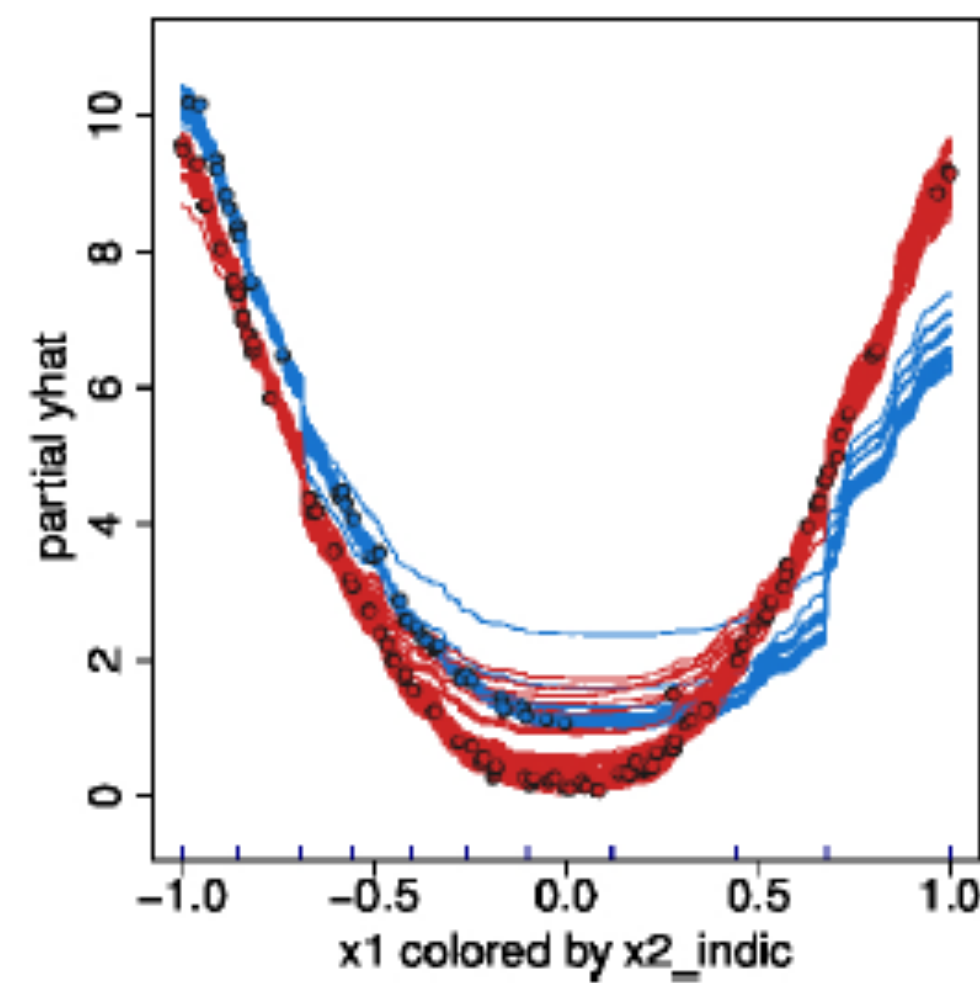
Consider data generated by the following model:

$$Y = 10X_1^2 + \mathbb{1}_{X_2 \geq 0} + \mathcal{E},$$

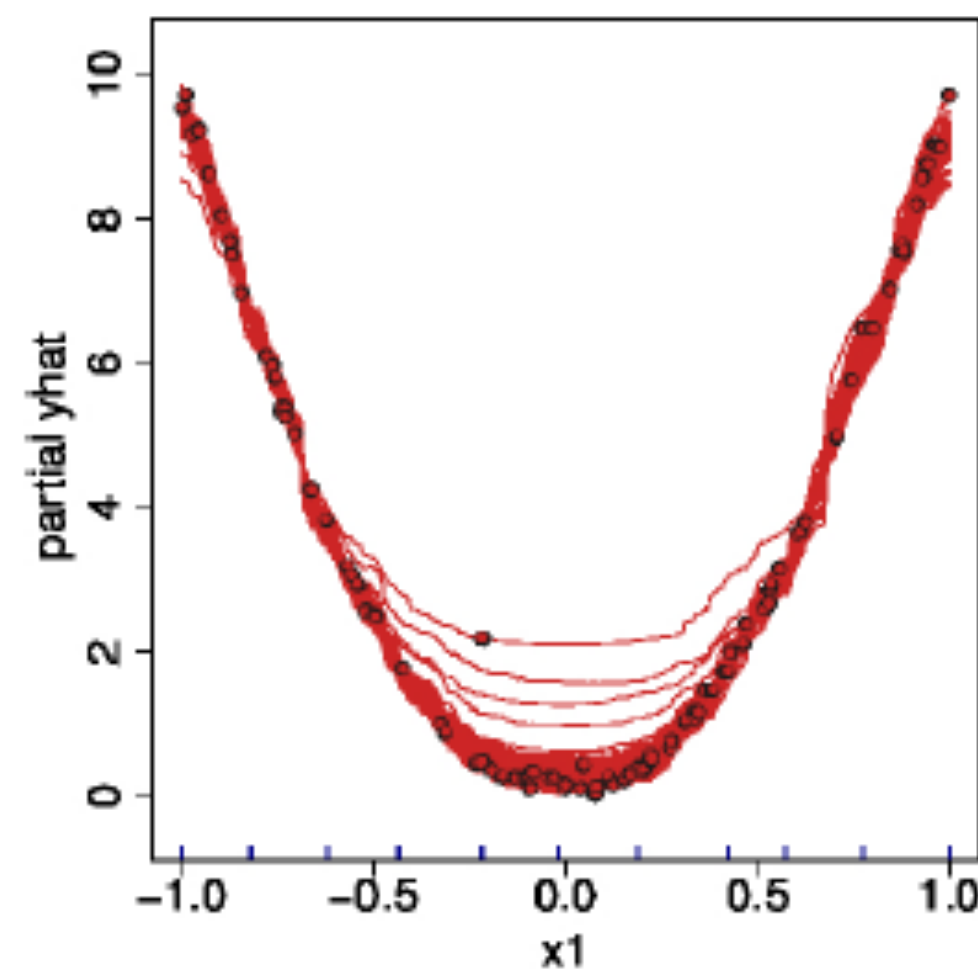
$$\mathcal{E} \stackrel{iid}{\sim} \mathcal{N}(0, .1^2), \quad \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} \sim \begin{cases} \text{U}(-1, 0), \text{U}(-1, 0) & \text{w.p. } \frac{1}{3} \\ \text{U}(0, 1), \text{U}(-1, 0) & \text{w.p. } \frac{1}{3} \\ \text{U}(-1, 0), \text{U}(0, 1) & \text{w.p. } \frac{1}{3} \\ \text{U}(0, 1), \text{U}(0, 1) & \text{w.p. } 0. \end{cases}$$

So that $P(X_1 > 0, X_2 > 0) = 0$.

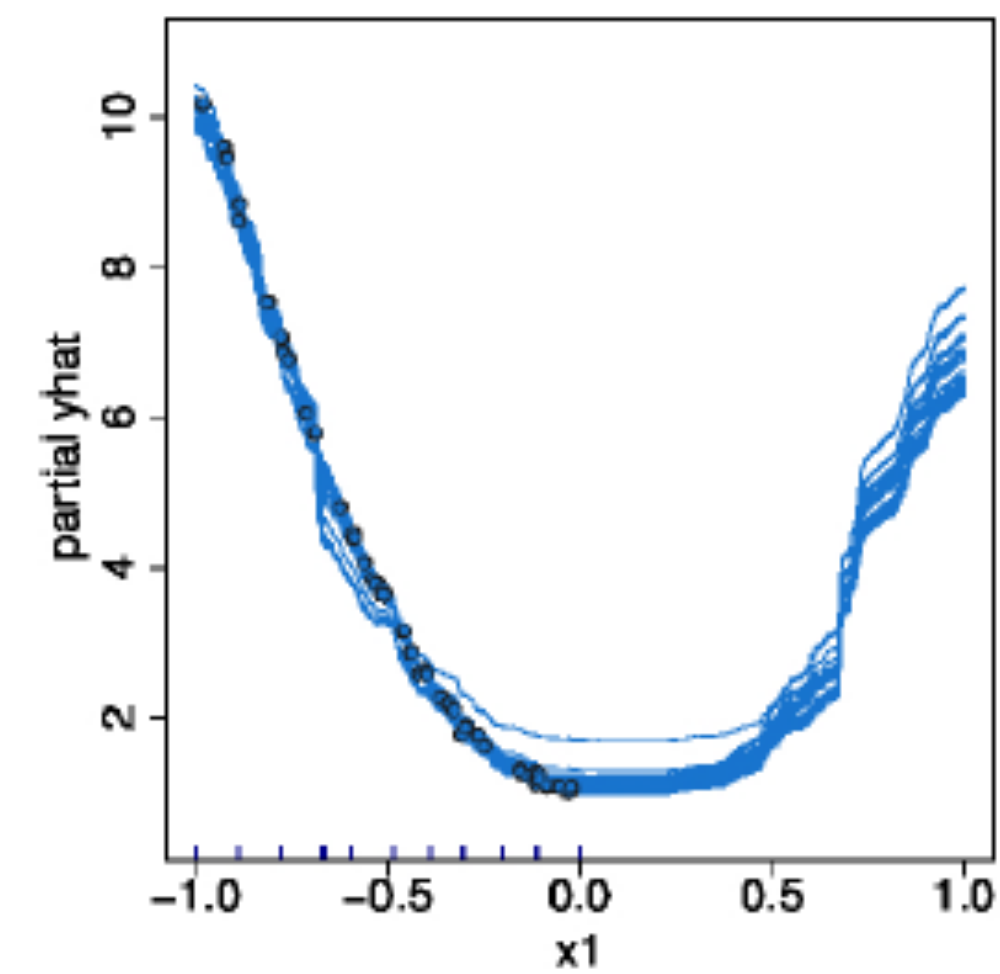
Uses of ICEs : additivity assessment



(a) All observations



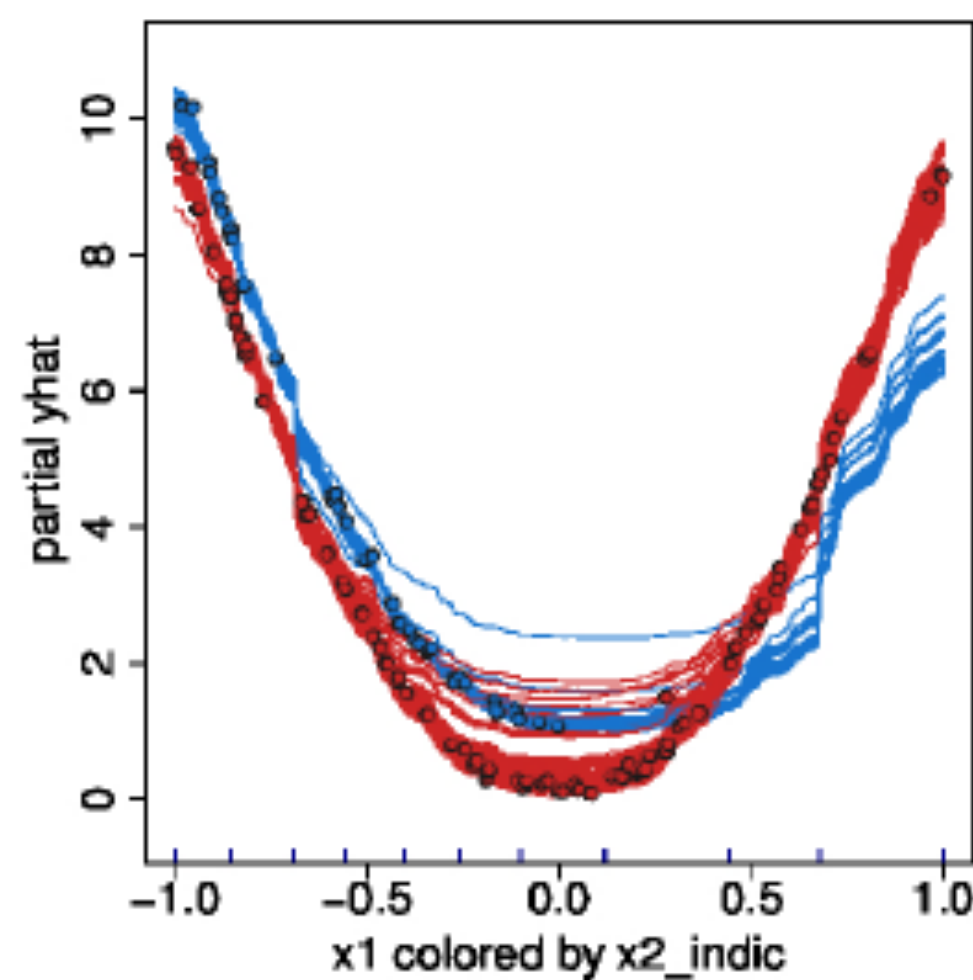
(b) Observations with $x_2 < 0$



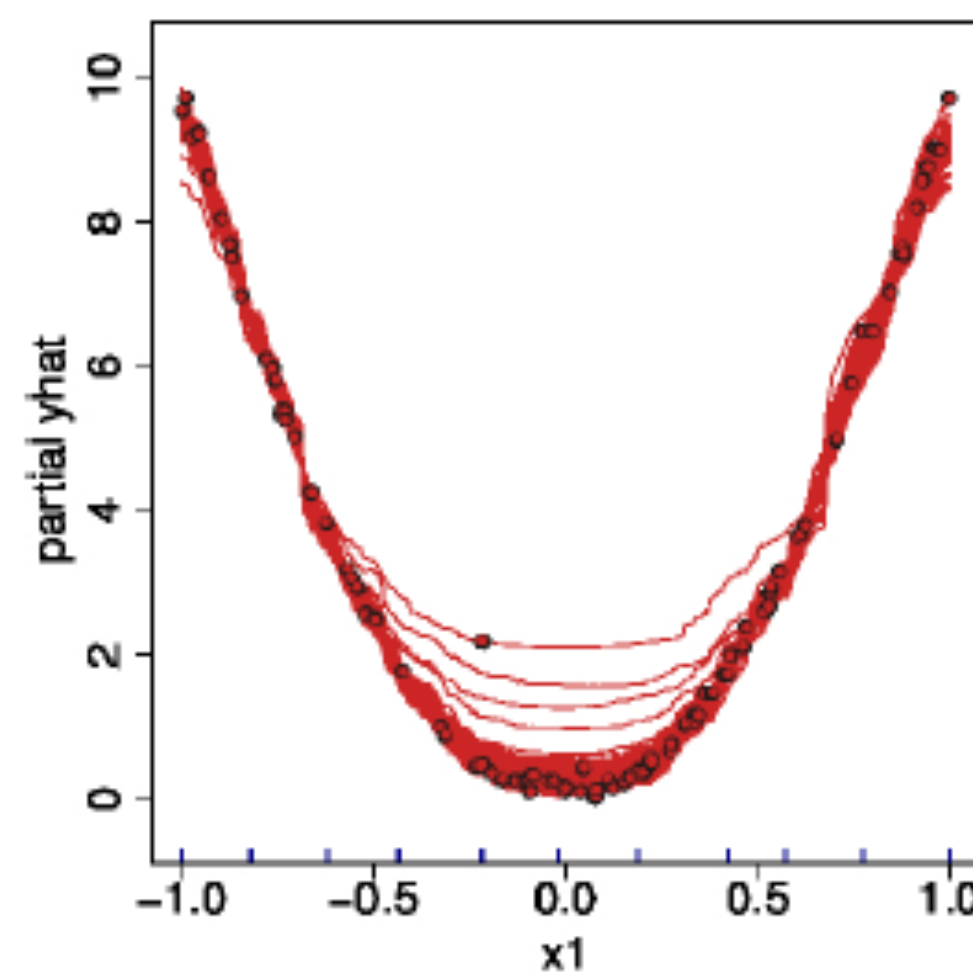
(c) Observations with $x_2 \geq 0$

Uses of ICEs : additivity assessment

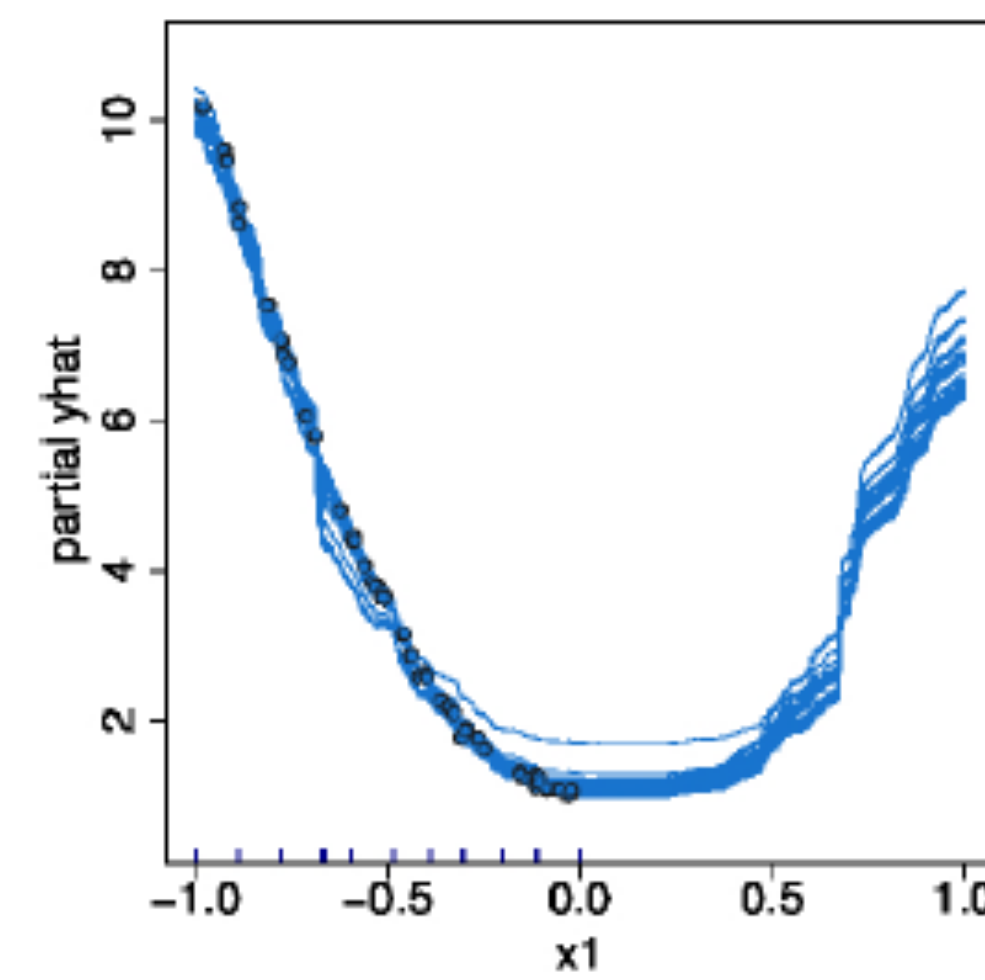
Here we show ICE plots w.r.t. X_1 for of a Random Forest model fitted on the generated data. The left plot shows the ICE plot for the entire dataset where $X_2 < 0$ is coloured in red and $X_2 \geq 0$ in blue. The middle plot shows only the red curves and the right only the blue. Recall that there is no training data in the quadrant $[0; 1] \times [0; 1]$. The blue ICE lines highlight how the model fit behaves in the unseen subspace, when both the variables are positive (it appears that the fitted values are increasing in X_1 for values above 0) .



(a) All observations



(b) Observations with $x_2 < 0$



(c) Observations with $x_2 \geq 0$

PDP and ICE – summing up

PDP and ICE – summing up

PDP and ICE are tools that can provide an intuitive explanation of how variables marginally affects the output of a black-box model without any assumption on the model itself.

ICE curves uncover heterogeneous relationship in the data. Often are used in combination with PDP to show also the average effect.

ICE can only visualize one variable meaningfully, while PDP can be used effectively also with two variables.

Both PDP and ICE suffer from the fact that they assume independence between x_l and the points $x_{i,-l}$, which is often unrealistic, leading to $F()$ being computed at points coming from low-probability regions.

Evaluation Metrics



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Evaluation Metrics - introduction

A. Rosenfeld. “Better Metrics for Evaluating Explainable Artificial Intelligence”, 2021

Evaluation Metrics - introduction

In the last few decades, the heterogeneity of XAI methods has grown more prominent.

Moreover, many explanations are generated post hoc, in a way that is independent of the model's internal processes.

User-based studies have become crucial for addressing the need for explanations, especially for non-experts, but these evaluations are highly subjective and depend on the application and target users.

To address these issues, Rosenfeld [3] tried to establish objective metrics to evaluate each explanation.

A. Rosenfeld. "Better Metrics for Evaluating Explainable Artificial Intelligence", 2021

Evaluation Metrics - introduction

Evaluation Metrics - introduction

Rosenfeld defines 4 objective metric to evaluate a black-box model explanations: ***D*** (difference), ***R*** (rules), ***F*** (features) and ***S*** (stability).

One of the advantages of these metrics is that being independent from the task at hand and the specific ML/DL algorithm being used, they cannot suffer from confirmation bias.

This may arise in user-based studies when the evaluation on the goodness of a method is purely based on human perception.

Evaluation Metrics - difference

Evaluation Metrics - difference

The metric **D** is defined as the difference between the black box model and the best transparent model with respect to a given measure of performance (*i.e.* MSE, accuracy, *et cetera*).

Let P_b the performance of the black-box model and P_t the performance of the transparent model. Compute

$$\delta = P_b - P_t$$

If δ is small enough, then the transparent model should be preferred. In practice, a threshold value for the metric should be chosen.

Evaluation Metrics - rules

Evaluation Metrics - rules

Following the Occam's razor principle, the idea behind the ***R*** metric is to prefer simple explanations. The fewer rules are used in the explanation the better.

Consider for example a decision tree, which is generally considered a transparent model. As the depth of the tree grows (adding rules to the decision process), the model becomes less interpretable.

In the literature it is suggested to penalize explanations with a number of rules above a certain threshold.

Evaluation Metrics - features

Evaluation Metrics - features

The ***F*** metric considers the number of variables used by the explanation method. Similarly to what said for the ***R*** metric, simpler explanation should be preferred.

Here, the intuition is that the less number of features are used, the easier the explanation is to interpret for a human user.

When computing this metric, one must take into account that at times simple features can be aggregated into more complex one to improve interpretability, *e.g.* single pixels in an image can be aggregated into edges.

Evaluation Metrics - stability

Evaluation Metrics - stability

The last metric **S** quantifies the stability of the agent's explanation with respect to small perturbation of the input feature.

In the literature, it is suggested to bootstrap the data and then observing its impact on the outputted agent explanations.

Only small perturbations should be used, as the performance of the model must not be affected.

Evaluation Metrics - summing up

Evaluation Metrics - summing up

Metrics' applicability varies by context and structural requirements; global methods may lack detail, affecting the feasibility of certain metrics such as *D*.

In general, it is easier to compare local methods and global methods separately since the scope of the explanation is vastly different. Depending on the structure of the explainer, it will be possible to provide a more accurate analysis or an intuition on the quality of the explanation itself.

The variety of XAI methods complicates comparisons, highlighting the need for in-depth analysis and human-based studies to discover effective, understandable methodologies.

THIS IS THE END



Thanks for the attention!